

BeamsEC2

Data	2026-07-16 16:54	Suporte normativo	NP EN 1992-1-1:2010
Envelopes	205	OK/Falhas	0/9

Resumo executivo

Indicador	Valor
Norma de cálculo	NP EN 1992-1-1
Vigas distintas	79.00
Envelopes analisados	205.00
Casos OK	0.00
Falhas ELU	9.00
Avisos ELS	0.00
Utilização global máxima	46.35
Resumo por viga	79.00

Resumo por viga

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
120 - V10.C	COBERTURA (CP)1	101 (C)	Rectangular	28.37	34.95	32.23	0.98	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
121 - Beam_121	PAVIMENTO TÉRREO	102 (C)	Rectangular	27.82	26.33	10.56	2.25	2012	2012	1010/face	Inf.: 2012; Sup.: 2012; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1010/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
122 - V1.C	COBERTURA (CP)1	106 (C)	Rectangular	127.97	79.57	88.95	19.77	2Ø20	2Ø20	1Ø16/face	Inf.: 2Ø20; Sup.: 2Ø20; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1Ø16/face	Verificar
125 - V9.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	78.38	54.03	0.04	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
136 - V6.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	78.55	42.03	2.99	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
139 - V1.1	PISO 1	101 (C)	Rectangular	0.00	151.38	169.02	46.09	2Ø20	5Ø20	3Ø16/face	Inf.: 2Ø20; Sup.: 5Ø20; Estribos fechados Ø10/2r // 20.0 cm; pele/alma: 3Ø16/face	Verificar
140 - V6.C	COBERTURA (CP)1	102 (C)	Rectangular	0.00	92.79	51.71	3.18	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Falha
141 - V6.C	COBERTURA (CP)1	103 (C)	Rectangular	0.00	46.07	29.13	6.34	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
142 - V6.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	132.84	61.04	0.78	2016	2020	1010/face	Inf.: 2016; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
171 - V3.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	198.81	96.88	1.91	2016	3020	1010/face	Inf.: 2016; Sup.: 3020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
172 - V3.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	100.43	44.10	0.16	2016	2020	1010/face	Inf.: 2016; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
173 - Beam_173	PAVIMENTO TÉRREO	103 (C)	Rectangular	6.66	7.60	17.01	5.15	2012	2012	1010/face	Inf.: 2012; Sup.: 2012; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1010/face	Verificar
174 - V5.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	23.87	22.56	7.19	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
175 - V5.C	COBERTURA (CP)1	105 (C)	Rectangular	0.00	26.84	25.29	4.70	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
176 - V5.C	COBERTURA (CP)1	105 (C)	Rectangular	8.43	24.44	23.42	2.04	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
177 - V5.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	37.29	25.77	16.97	2Ø16	2Ø16	1Ø16/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 1Ø16/face	Verificar
178 - V8.C	COBERTURA (CP)1	106 (C)	Rectangular	0.00	27.84	22.51	0.15	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
179 - V8.C	COBERTURA (CP)1	105 (C)	Rectangular	0.00	31.25	23.09	0.09	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
180 - V8.C	COBERTURA (CP)1	105 (C)	Rectangular	0.00	32.58	23.31	0.98	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
181 - V8.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	30.67	23.59	0.22	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
182 - V8.C	COBERTURA (CP)1	102 (C)	Rectangular	0.00	43.21	25.81	3.57	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
186 - V1.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	169.24	80.17	13.78	2Ø20	2Ø20	1Ø12/face	Inf.: 2Ø20; Sup.: 2Ø20; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1Ø12/face	Verificar
188 - V4.C	COBERTURA (CP)1	107 (C)	Rectangular	0.00	26.88	23.49	8.99	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
189 - V7.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	26.99	29.70	2.84	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
196 - Beam_196	PAVIMENTO TÉRREO	102 (C)	Rectangular	0.00	14.99	541.13	6.25	2Ø12	2Ø12	1Ø10/face	Inf.: 2Ø12; Sup.: 2Ø12; Estribos fechados Ø10/6r // 12.5 cm; pele/alma: 1Ø10/face	Falha
201 - V4.C	COBERTURA (CP)1	105 (C)	Rectangular	0.00	36.67	29.83	0.03	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
210 - V1.C	COBERTURA (CP)1	101 (C)	Rectangular	144.96	171.96	110.93	8.31	3Ø20	3Ø20	1Ø16/face	Inf.: 3Ø20; Sup.: 3Ø20; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø16/face	Verificar
225 - V6.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	35.64	30.09	3.11	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
226 - V6.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	52.83	38.72	3.20	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
268 - V1.1	PISO 1	101 (C)	Rectangular	0.00	90.74	100.71	25.98	2Ø16	3Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 3Ø20; Estribos fechados Ø8/2r // 20.0 cm; pele/alma: 2Ø16/face	Verificar
276 - V1.1	PISO 1	101 (C)	Rectangular	0.00	188.58	199.59	35.29	2Ø16	5Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 5Ø20; Estribos fechados Ø10/2r // 20.0 cm; pele/alma: 2Ø16/face	Verificar
278 - V1.1	PISO 1	101 (C)	Rectangular	0.00	134.27	105.94	2.68	2Ø12	3Ø20	1Ø10/face	Inf.: 2Ø12; Sup.: 3Ø20; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
Viga 283	PAVIMENTO TÉRREO	102 (C)	Rectangular	59.64	0.00	6999	8.90	2020	2012	1012/face	Inf.: 2020; Sup.: 2012; Estribos fechados Ø10/6r // 7.5 cm; pele/alma: 1012/face	Falha
284 - V1.1	PISO 1	102 (C)	Rectangular	10.16	112.91	51.14	6.41	2012	3020	1010/face	Inf.: 2012; Sup.: 3020; Estribos fechados Ø8/2r // 20.0 cm; pele/alma: 1010/face	Verificar
Viga 285	PAVIMENTO TÉRREO	102 (C)	Rectangular	2.57	1.26	7.51	16.34	2012	2012	1016/face	Inf.: 2012; Sup.: 2012; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1016/face	Verificar
Viga 306	PAVIMENTO TÉRREO	101 (C)	Rectangular	8.23	41.52	61.31	3.27	2012	2016	1010/face	Inf.: 2012; Sup.: 2016; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1010/face	Verificar
Viga 307	PAVIMENTO TÉRREO	103 (C)	Rectangular	0.00	61.19	103.33	45.03	2020	3020	3016/face	Inf.: 2020; Sup.: 3020; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 3016/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
308 - V1.1	PISO 1	102 (C)	Rectangular	0.00	41.38	38.00	3.49	2012	2016	1010/face	Inf.: 2012; Sup.: 2016; Estribos fechados Ø8/2r // 20.0 cm; pele/alma: 1010/face	Verificar
309 - V2.1	PISO 1	104 (C)	Rectangular	0.00	62.16	51.08	0.74	2020	2020	1010/face	Inf.: 2020; Sup.: 2020; Estribos Ø10/2r // 25.0 cm; pele/alma: 1010/face	Verificar
310 - V2.1	PISO 1	105 (C)	Rectangular	0.00	60.62	39.12	2.92	2020	2020	1010/face	Inf.: 2020; Sup.: 2020; Estribos Ø10/2r // 25.0 cm; pele/alma: 1010/face	Verificar
312 - V4.C	COBERTURA (CP)1	109 (C)	Rectangular	0.00	38.88	29.99	1.63	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
313 - V1.1	PISO 1	101 (C)	Rectangular	0.00	119.08	29205	169.26	3020	5020	4016/face	Inf.: 3020; Sup.: 5020; Estribos fechados Ø10/6r // 7.5 cm; pele/alma: 4016/face	Falha
315 - V1.1	PISO 1	101 (C)	Rectangular	0.00	71.54	30.65	0.59	2012	2016	1010/face	Inf.: 2012; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
316 - V1.1	PISO 1	104 (C)	Rectangular	0.00	49.34	42.99	4.09	2Ø12	2Ø16	1Ø10/face	Inf.: 2Ø12; Sup.: 2Ø16; Estribos fechados Ø8/2r // 20.0 cm; pele/alma: 1Ø10/face	Verificar
322 - V4.C	COBERTURA (CP)1	112 (C)	Rectangular	0.00	36.16	29.91	0.72	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
327 - V1.1	PISO 1	105 (C)	Rectangular	0.00	201.91	213.44	81.66	2Ø16	5Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 5Ø20; Estribos fechados Ø10/4r // 20.0 cm; pele/alma: 2Ø16/face	Verificar
330 - V1.1	PISO 1	101 (C)	Rectangular	0.00	131.98	108.44	72.78	2Ø20	4Ø20	3Ø16/face	Inf.: 2Ø20; Sup.: 4Ø20; Estribos fechados Ø8/4r // 20.0 cm; pele/alma: 3Ø16/face	Verificar
360 - V1.1	PISO 1	101 (C)	Rectangular	0.00	94.00	137.09	102.08	2Ø20	3Ø20	3Ø16/face	Inf.: 2Ø20; Sup.: 3Ø20; Estribos fechados Ø10/6r // 20.0 cm; pele/alma: 3Ø16/face	Falha
365 - V1.2	PISO TÉCNICO	102 (C)	Rectangular	0.00	55.67	42.14	6.71	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
366 - V3.2	PISO TÉCNICO	104 (C)	Rectangular	0.00	120.82	187.82	3.10	2016	2020	1010/face	Inf.: 2016; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
367 - V3.2	PISO TÉCNICO	102 (C)	Rectangular	0.00	255.96	348.32	68.36	2020	6020	3016/face	Inf.: 2020; Sup.: 6020; Estribos fechados Ø10/4r // 25.0 cm; pele/alma: 3016/face	Verificar
368 - V3.2	PISO TÉCNICO	102 (C)	Rectangular	0.00	176.54	185.36	40.16	2016	4020	2016/face	Inf.: 2016; Sup.: 4020; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 2016/face	Verificar
369 - V1.2	PISO TÉCNICO	102 (C)	Rectangular	0.00	111.58	87.46	27.10	2016	3020	2012/face	Inf.: 2016; Sup.: 3020; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 2012/face	Verificar
370 - V3.2	PISO TÉCNICO	101 (C)	Rectangular	0.00	199.21	225.50	53.69	2020	5020	3016/face	Inf.: 2020; Sup.: 5020; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 3016/face	Verificar
371 - V3.2	PISO TÉCNICO	104 (C)	Rectangular	0.00	261.60	316.32	39.38	2016	5020	2016/face	Inf.: 2016; Sup.: 5020; Estribos fechados Ø8/4r // 25.0 cm; pele/alma: 2016/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
372 - V3.2	PISO TÉCNICO	104 (C)	Rectangular	0.00	135.45	197.11	9.61	2Ø16	3Ø20	1Ø10/face	Inf.: 2Ø16; Sup.: 3Ø20; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
375 - V2.2	PISO TÉCNICO	101 (C)	Rectangular	26.44	29.53	42.57	1.41	2Ø16	2Ø16	n/a	Inf.: 2Ø16; Sup.: 2Ø16; Estribos fechados Ø8/2r // 12.5 cm	Verificar
Viga 377	PAVIMENTO TÉRREO	102 (C)	Rectangular	5.33	93.21	153.96	3.73	2Ø12	2Ø20	1Ø10/face	Inf.: 2Ø12; Sup.: 2Ø20; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1Ø10/face	Verificar
378 - V4.C	COBERTURA (CP)1	112 (C)	Rectangular	0.00	26.56	23.51	0.62	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
380 - Viga 21 (I-140)	COBERTURA (CP)1	101 (C)	I asimétrica	0.00	0.00	231.85	0.00	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
381 - Viga 22 (I-140)	COBERTURA (CP)1	104 (C)	I asimétrica	0.00	0.00	99.29	0.00	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
382 - Viga 23 (I-140)	COBERTURA (CP)1	104 (C)	I asimétrica	0.00	0.00	121.46	0.00	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
383 - Viga 24 (I-140)	COBERTURA (CP)1	104 (C)	I asimétrica	0.00	0.00	134.03	0.00	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
392 - V4.C	COBERTURA (CP)1	108 (C)	Rectangular	0.00	40.63	26.51	3.55	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
396 - V4.C	COBERTURA (CP)1	111 (C)	Rectangular	0.00	55.19	38.89	7.15	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos fechados Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
439 - V10.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	29.46	24.17	15.99	2016	2016	1016/face	Inf.: 2016; Sup.: 2016; Estribos fechados Ø10/2r // 25.0 cm; pele/alma: 1016/face	Verificar
440 - V10.C	COBERTURA (CP)1	101 (C)	Rectangular	28.37	61.82	72.17	0.97	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
457 - V3.C	COBERTURA (CP)1	104 (C)	Rectangular	0.00	107.40	73.40	3.33	2016	2020	1010/face	Inf.: 2016; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
459 - V1.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	121.74	67.24	9.90	2020	2020	1012/face	Inf.: 2020; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1012/face	Verificar
462 - V1.C	COBERTURA (CP)1	101 (C)	Rectangular	0.00	130.59	90.80	3.21	2020	2020	1012/face	Inf.: 2020; Sup.: 2020; Estribos Ø8/2r // 25.0 cm; pele/alma: 1012/face	Verificar
463 - V10.C	COBERTURA (CP)1	106 (C)	Rectangular	10.94	26.85	24.65	1.48	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Verificar
47 - V6.C	COBERTURA (CP)1	106 (C)	Rectangular	0.00	87.50	48.73	2.10	2016	2016	1010/face	Inf.: 2016; Sup.: 2016; Estribos Ø8/2r // 25.0 cm; pele/alma: 1010/face	Falha
489 - Beam_489	PAVIMENTO TÉRREO	102 (C)	Rectangular	0.00	39.38	99.82	8.08	2012	2016	1012/face	Inf.: 2012; Sup.: 2016; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1012/face	Verificar

viga	story	case	section_type	m_pos_ed_kNm	m_neg_ed_kNm	v_ed_kN	t_ed_kNm	bot_rebar	top_rebar	skin_rebar	solution	status
491 - Beam_491	PAVIMENTO TÉRREO	101 (C)	Rectangular	0.00	163.59	202.56	32.82	2Ø16	5Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 5Ø20; Estribos fechados Ø10/2r // 17.5 cm; pele/alma: 2Ø16/face	Verificar
492 - Beam_492	PAVIMENTO TÉRREO	101 (C)	Rectangular	0.00	66.74	62.14	2.37	2Ø12	2Ø16	1Ø10/face	Inf.: 2Ø12; Sup.: 2Ø16; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 1Ø10/face	Verificar
493 - Beam_493	PAVIMENTO TÉRREO	101 (C)	Rectangular	0.00	85.99	116.71	31.83	2Ø16	3Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 3Ø20; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 2Ø16/face	Verificar
494 - Beam_494	PAVIMENTO TÉRREO	105 (C)	Rectangular	0.00	63.48	63.42	30.61	2Ø16	3Ø20	2Ø16/face	Inf.: 2Ø16; Sup.: 3Ø20; Estribos fechados Ø8/2r // 17.5 cm; pele/alma: 2Ø16/face	Verificar
6 - V4.C	COBERTURA (CP)1	302 (C)	Rectangular	0.00	26.67	22.63	0.10	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar
7 - V4.C	COBERTURA (CP)1	109 (C)	Rectangular	0.00	39.77	30.70	0.83	2Ø16	2Ø16	1Ø10/face	Inf.: 2Ø16; Sup.: 2Ø16; Estribos Ø8/2r // 25.0 cm; pele/alma: 1Ø10/face	Verificar